CLASS 9 · ICSE · SELINA CONCISE MATHEMATICS · DPP SHEET

Chapter 21: Solids

Daily practice problems: 10 questions with an answer key

Try every question on your own first. The answer key is on the last page.

1. Find the volume and total surface area of a cuboid 10 cm by 8 cm by 5 cm.
2. Find the length of the longest rod that can be placed in a room 12 m by 9 m by 8 m.
3. The volume of a cube is 512 cu cm. Find its total surface area.
4. The diagonal of a cube is $6\sqrt{3}$ cm. Find its volume.
5. How many bricks, each 25 cm by 10 cm by 8 cm, are needed for a wall 5 m long, 2 m high and 0.4 m thick?
6. A tank 2.5 m by 2 m by 1.2 m is full of water. How many litres does it hold?
7. An open box is 20 cm long, 15 cm wide and 10 cm high (external). Find its outer surface area.
8. A prism has a cross-section of area 35 sq cm and length 12 cm. Find its volume.
9. Water flows at 3 m/s through a pipe whose cross-section is 20 sq cm. How many litres flow in 10 seconds?
10. The edge of a cube is doubled. By what factor do its volume and surface area change?

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Answer Key

1. 400 cu cm; 340 sq cm
2. 17 m
3. Edge 8 cm; 384 sq cm
4. 216 cu cm
5. 2000
6. 6000 litres
7. $300 + 700 = 1000$ sq cm
8. 420 cu cm
9. 60 litres
10. Volume 8 times; surface area 4 times

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